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Key Points:

- Global Sporadic E (Es) layers exhibited significant yet distinct disturbance responses to the May 2024 storm
- Signs of enhanced coupling between the Es and F regions have likely been revealed by simultaneous radio occultation observations
- Wave-like propagation from the polar to the equatorial region was identified in the Southern Hemisphere

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Analysis on Global Es Layer Response to the May 2024 Geomagnetic Storm Through a Combination of Tianmu-1 and COSMIC-2 Radio Occultation Observations

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Abstract The sporadic E (Es) layer, a key manifestation of ionosphere-neutral atmosphere coupling, is highly sensitive to geomagnetic storms. Using more than 25,000 daily radio occultation profiles from the Tianmu-1 and COSMIC-2 constellations, this study investigates the global response of the Es layer to the May 2024 storm. Observations reveal that the storm produced globally extensive disturbances in the Es region, exhibiting distinct latitudinal, longitudinal, and storm-phase dependences. During the main phase, the peak Es-layer intensity (EsI) exceeded four times the quiet-time level at high latitudes, and the Es-layer height (EsH) rose by as much as 20 km. Simultaneously, EsI at mid and low latitudes showed depletions exceeding 50%. During the recovery phase, wave-like equatorward propagations of enhanced EsI and elevated EsH were observed in the Southern Hemisphere, primarily over the Southeast Pacific sector. Nighttime co-occurrence rates of Es- and F-region irregularities increased by a factor of four, suggesting that F-region ionospheric perturbations may also contribute to Es-layer generation during the recovery phase. These findings provide observational evidence of the global impact of a G5-class geomagnetic storm on the Es layer, offering valuable insights for the development and validation of models that predict Es-layer behavior.

Plain Language Summary The Chinese Tianmu-1 commercial satellite constellation has provided a large number of ionospheric RO profiles, for the first time, enabling a global analysis of the Es layer behavior during a G5-level geomagnetic storm in May 2024. Analysis reveals that the global Es layer was significantly disturbed during the storm, with distinct latitudinal and longitudinal behaviors. Based on observational evidence, potential mechanisms for the formation of the storm-time Es layer are indicated in this study. First, particle precipitation likely caused the enhancement of Es intensity (EsI) at high latitudes, while the intense eastward prompt penetration electric field may be the primary driver for mid-to-low-latitudes EsI depletion during the main phase. Second, storm-driven neutral winds may have contributed to Es layer development over the Australia-Asia and Southeast Pacific sectors in the Southern Hemisphere. Third, F-region ionosphere perturbations may exert downward influences on the underlying Es region.

1. Introduction

The May 2024 storm stands as the first G5-class geomagnetic storm in the past two decades, following the Halloween geomagnetic storm in 2003 (Evans et al., 2024). Also referred to as the Gannon storm or Mother's Day storm, it is hereafter termed as the “May 2024 storm.” During this event, enhanced auroral particle precipitation and electric fields drove substantial Joule heating and ion-neutral drag at high latitudes, leading to pronounced global disturbances in neutral composition (Liu et al., 2025), thermospheric winds (Wang et al., 2025), and F-region ionosphere (Du and Yang, 2025; Sun et al., 2025; Themens et al., 2024; Yang et al., 2025; Zhang et al., 2024).

Sporadic E layers (Es layers) are thin-layered structures within the ionospheric E region at altitudes of 90–140 km. They are characterized by electron densities three to four orders of magnitude higher than the background ionosphere and occasionally exceed the F-layer peak density (He et al., 2024; Sheng et al., 2025; Tian et al., 2023; Tian et al., 2025). The Es layer is modulated by complex drivers such as lower atmospheric forcing, ionospheric processes, and solar activity (Andoh S et al., 2022; Tang et al., 2020; Sun et al., 2025; Yamazaki et al., 2021). The local time and seasonal variations of Es layers during geomagnetic quiet conditions have been well documented, mainly through Global Navigation Satellite System Radio Occultation (GNSS RO) observations.

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Several studies have examined Es-layer responses to geomagnetic storms in low-latitude regions (e.g., Abdu et al., 2013; Moro et al., 2022; Resende et al., 2024; Zhang et al., 2015). Geomagnetic activity has been shown to account for approximately 49% of the variability in nighttime E-region electron density (Salinas et al., 2025). In extreme cases, such as the 2003 Halloween storm, reverse electrojet currents produced a 146% enhancement in Es-layer electron density (Clezio et al., 2016). Qiu and Lin. (2025a) and Chang et al. (2025a, 2025b) reported the Es-layer response during the May 2024 storm using ground-based observations and a limited set of RO measurements. However, the global morphology of Es layers during a G5-class event like the May 2024 storm remains poorly characterized.

The E region not only exhibits its own dynamic behavior but also plays a crucial role in the vertical redistribution of ionospheric plasma through coupling with the F region. Proposed mechanisms for this coupling include the seeding of F-region irregularities by gravity waves originating in the Es layer (Joshi, 2016; Joshi et al., 2019; Patra et al., 2005), as well as modulation of Es layers by polarization electric fields from F-region irregularities (Joshi et al., 2013). Statistical studies confirm a strong correlation between Es and F-region activity (Li et al., 2011, 2023; Patra et al., 2014; Xie et al., 2018), with the strength of this coupling partially quantified by the co-occurrence rates of irregularities in both regions (Liu et al., 2019, 2020; Yang and Liu, 2018). Therefore, understanding the response of the E region to such an extreme storm is essential for advancing our knowledge of ionospheric mechanisms during extreme space weather events.

Recent advancements in commercial RO constellations have greatly enriched the research on the storm-time ionosphere (Lee et al., 2025). China's Tianmu-1 constellation comprises 23 low-earth-orbit (LEO) satellites which can provide over 20,000 ionospheric RO profiles daily (Yang and Zou, 2025). Combined with observations from the second Constellation Observing System for Meteorology, Ionosphere, and Climate satellites (COSMIC-2) mission (Ho et al., 2020; Schreiner et al., 2020), over 25,000 RO profiles enable, for the first time, a global analysis of Es layer responses to a G5-level storm. Unlike previous research that relied on single tangent point data, this study utilizes full RO profiles to simultaneously examine Es-region irregularities and F-region irregularities. Meanwhile, F-region irregularities detected by over 2,600 ground-based GNSS stations are also illustrated.

2. Data and Method

Figure 1 shows the geomagnetic indices for the May 2024 storm. Following Halford et al. (2010), the main phase begins when the SYM-H index turns negative and persists until the SYM-H index reaches its minimum value. The recovery phase extends from this minimum until the SYM-H index recovers to 80% of its minimum value relative to the pre-storm baseline. Storm onset occurred at 17:54 UT on 10 May 2024, when the SYM-H index became negative. The main phase extended from 17:54 on 10 May to 02:14 UT on 11 May during which SYM-H index decreased from -7 nT to a minimum of -512 nT. The Kp index reached a maximum value of 9, confirming the classification of the May 2024 event as an extreme G5-class geomagnetic storm by the NOAA Space Weather Prediction Center (Evans et al., 2024). A sharp increase in the AE index, accompanied by a concurrent rise in the AU index, indicated significant intensification of the westward auroral electrojet, this marked the expansion phase of substorm activity. The recovery phase then lasted from 02:14 UT 11 May to 12:00 UT 12 May, as the SYM-H index recovered to -105 nT (calculated as: $(1\%-80\%) \times [-512 - (-7)]$ nT = -103.8 nT).

The Es intensity (EsI), Es perturbation (EsP), Es height (EsH), Es height perturbation (EsHP) and Es region irregularities (EIr) were adopted to characterize the Es layer response to the May 2024 storm. The EsI is defined as the maximum amplitude scintillation index S4 within the 90–140 km altitude range. The EsP is calculated as $\frac{EsI_{\text{during the storm}} - EsI_{\text{reference value}}}{EsI_{\text{reference value}}}$, where $EsI_{\text{reference value}}$ is the average EsI during 7–8 May and 14–15 May. The S4 profiles were obtained from 100-Hz signal-to-noise-ratio data (Zeng and Sokolovskiy, 2010) collected by the COSMIC-2 and Tianmu-1 RO measurements. EsH refers to the altitude at which the EsI is observed, and EsHP is defined as the difference between EsH during the storm and its quiet-time value. EIr was identified if EsI exceeded a threshold of 0.3. During data quality control, profiles with S4 values exceeding 3, a maximum altitude below 120 km, or a minimum altitude above 90 km were removed.

The geolocation for each RO observation was assigned to the tangent point of the line of sight between the GNSS and LEO satellites (Guo et al., 2024), a method widely used in previous research (e.g., Du and Yang, 2025;

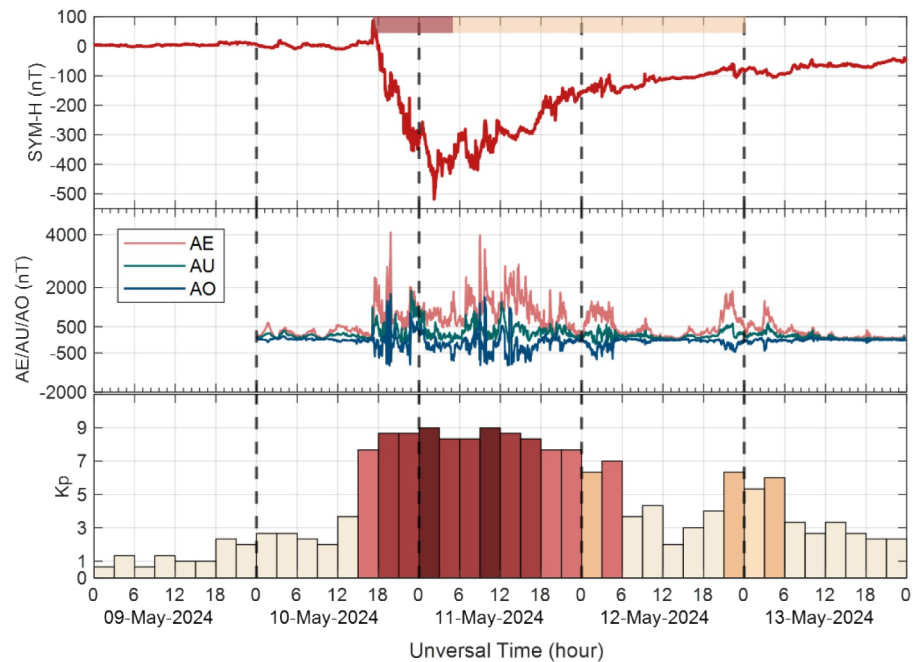


Figure 1. Time series of the geomagnetic indices during May 2024 storm. (top panel) the SYM-H index, representing the intensity of the symmetrical ring current; (middle panel) the AE/AU/AO indices, monitoring auroral electrojet activity associated with polar substorms; (low panel) the Kp index, quantifying global geomagnetic disturbance levels.

Dymond, 2012; Liu et al., 2020; Yang and Liu, 2018). In a RO event, the point along the signal path from the GPS to the LEO that is closest to the center of the Earth is defined as the tangent point (Guo et al., 2025a, 2025b).

It should be noted that inherent uncertainty exists in determining the exact location of each irregularity measurement along the RO plane, as each scintillation observation represents an integrated quantity (e.g., Carter et al., 2013; Sokolovskiy et al., 2002). Extreme cases with significant location uncertainty typically arise when the tangent point altitude is considerably lower than the peak of the F-layer. To mitigate this, profiles were excluded if the altitude difference between the tangent point and the F-layer peak exceeded 200 km, or if the horizontal distance between the tangent point calculated using the full profile and that calculated using the 200–500 km altitude segment differed by more than 2°.

Multiple RO events are aggregated by location and time, and that statistical occurrence trends are employed to infer underlying ionospheric processes. Straus et al. (2003) demonstrated that GNSS signal scintillation predominantly occurs near the tangent point of the line-of-sight, supporting the validity of using the tangent point to locate ionospheric irregularities. Thus, E- and F-region irregularities detected within the same RO profile are assigned at the tangent point in this study.

The standard deviation of the rate of total electron content (TEC) change (ROTI) was used to characterize the F-region irregularities. ROTI data were derived from two sources: 1-s TEC measurements from RO observations and 30-s TEC measurements from over 2600 ground-based GNSS stations. The RO technique provides ROTI profiles from 60 km altitude up to the LEO satellite orbit (approximately 500 km for Tianmu-1 and COSMIC-2), and its feasibility for monitoring F-region irregularities has been validated (e.g., Liu et al., 2020; Sokolovskiy et al., 2024; Yang and Liu, 2018). For the ground-based GNSS data, a thin shell ionosphere model at 350 km was used to calculate the ionospheric pierce point, with an elevation angle higher than 60° applied to remove multipath effects. ROTI was calculated over a 5-min window for ground-based GNSS observations and a 5-s window for the short-duration RO events. When ROTI exceeded 0.15 TECU/s above 200 km, F-region irregularities were considered to have occurred (Liu et al., 2020).

Figure 2 depicts the global distribution of RO tangent points and ground-based GNSS stations on 11 May 2024. The left panel underscores RO events from the Tianmu-1 constellation, which provided over 21,700 RO profiles per day with global coverage, while COSMIC-2's primary focus was on mid and low latitudes. The right panel

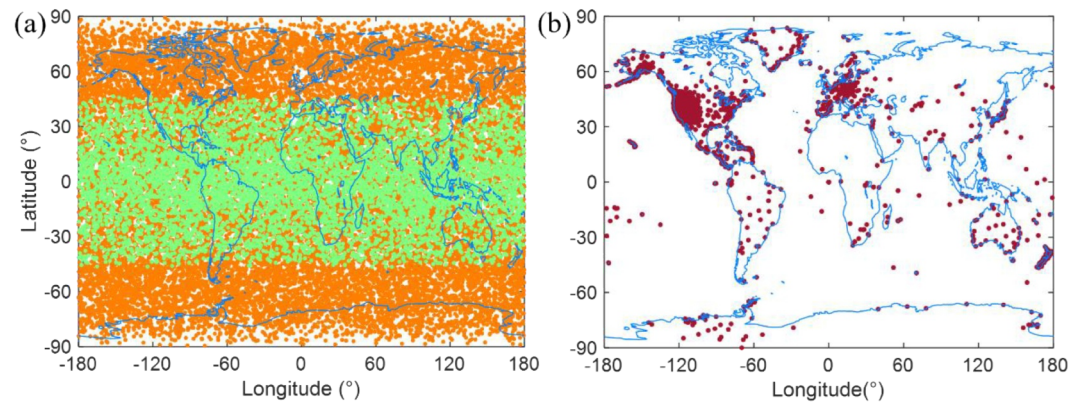


Figure 2. The geographical distribution of tangent point from 38,428 RO events (a) and 2761 ground-based GNSS stations (b) on 11 May 2024. The orange, green, and red dots denoted Tianmu-1 RO, COSMIC-2 RO and station positions, respectively.

displays the distribution of 2761 ground GNSS stations, with primary coverage extending across North and South America, Europe, Asia, and Australia.

3. Results

3.1. Global Morphology of Es Layer

Figure 3 presents the global distributions of EsI and EsP derived from RO observations during the magnetic quiet time, the storm main phase, and the recovery phase, respectively. Figures 3a and 3d show EsI and EsP under magnetic quiet conditions. EsI values over 0.3 occurred predominantly at the mid-latitudes in the Northern Hemisphere. This spatial distribution aligns with the known seasonal and regional characteristics of Es layers, which typically peak at middle latitudes during summer (Hu et al., 2022).

Figures 3b and 3c display EsI and EsP during the storm main phase. Both hemispheres showed strong EsI enhancement at high magnetic latitudes (MLAT) above 70°. The peak EsI values reached 0.6, about four times the quiet-time level, while the corresponding EsP peaks rose to 300%. In mid- and low- MLAT regions, EsI in the Eastern Hemisphere remained comparable to quiet-time values, whereas the Es layers in the Western Hemisphere were notably disrupted.

Figures 3e and 3f present results during the storm recovery phase. Es activity weakened but persisted at high latitudes, while enhanced EsI extended to mid- and low-latitudes, with EsI maxima exceeding 0.4 and EsP values around 80%.

To further examine the temporal evolution of the Es layer in response to the May 2024 storm, Figures 3g–3j show time series (in both UT and local time, LT) of EsI and EsP as a function of MLAT from May 8 to May 13. Before May 10, EsI above 0.2 primarily occurred between 30°N and 60°N MLAT, with the peak value appearing between 6:00 LT and 14:00 LT, and 18:00 LT and 24:00 LT. This quiet-time Es layer morphology was consistent with statistical results from previous studies (e.g., Niu et al., 2019; Yang and Liu, 2018). At high-latitudes, EsI enhancement began around 17:00 UT on May 10, following the storm onset, and persisted until May 12. At mid- and low-latitudes, EsI began to increase around 11:00 UT on May 11, however, EsI was significantly weakened between 17:00 UT on May 10 and 06:00 UT on May 11, falling below 0.1, while EsP reached a minimum of ~50%.

Figure 4 shows the global distributions of EsH and its perturbation. In the primary quiet-time Es occurrence region (30°N–70°N MLAT; see Figures 3a, 3d, and 3i), EsH typically ranged between 100 and 105 km as shown in Figures 4a, 4d, and 4i. During the storm, regions of enhanced EsI were accompanied by a marked uplift in EsH. During the main phase, EsH in high-latitude areas exceeded 120 km with a maximum of about 140 km, while corresponding EsHP values reached up to 20 km as shown in Figures 4b and 4c. During the recovery phase, EsH over the Southeast Pacific (90°W–150°W) and Australia-Asia sectors (120°E–180°E) lay between 115 and 120 km, accompanied by EsHP variations on the order of about 5–10 km, as shown in Figures 4e and 4f.

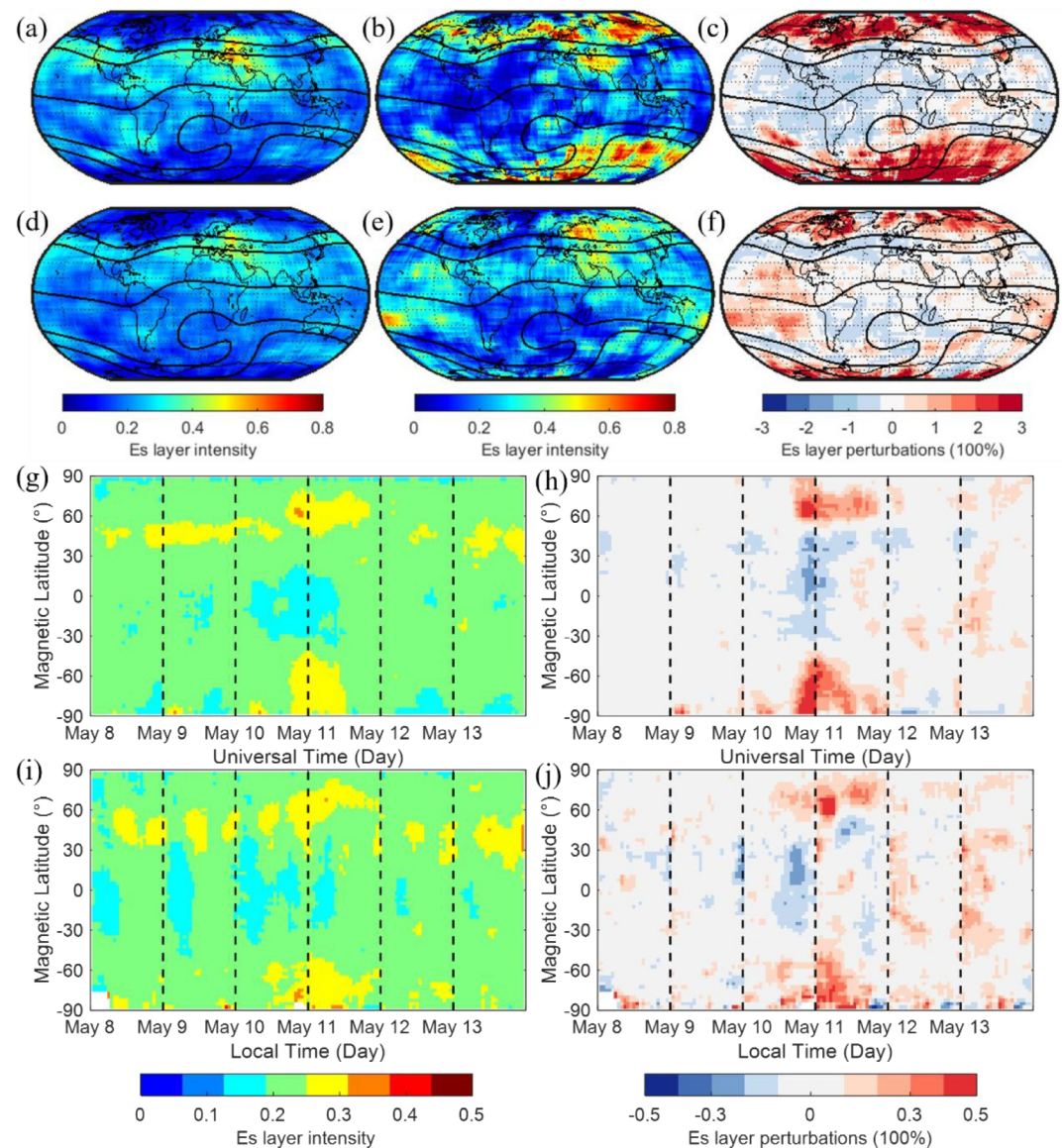


Figure 3. Geographical and temporal distributions of EsI and EsP derived from RO observations. The EsI (a, b, d, and e) and EsP (c, f) are plotted on a grid with a resolution of 5° in latitude and 5° in longitude. (a) EsI during the quiet-time reference period (17:00–00:00 UT); (b) EsI during the storm main phase; (d) EsI during the quiet-time reference period (02:00–17:00 UT); and (e) EsI during the storm recovery phase. (c) EsP during the main phase, (f) EsP during the recovery phase. (g and h) show the time series (in UT) of EsI and EsP, as a function of latitude from 8 to 14 May. (i and j) are the same as (g and h) but expressed in local time. In panels (a–f), the solid black lines denote magnetic latitude contours at -70° , -60° , 0° , 60° , and 70° , respectively.

Figures 4g and 4h present the time series (in UT) of EsH and EsHP. An increase in EsH was observed following the storm onset around 17:00 UT on 10 May. This height elevation exhibited a propagation from polar to equatorial regions: EsHP values exceeding 5 km were detected in polar areas at 16:30 UT on 10 May, reaching equatorial latitudes by approximately 12:00 UT on 11 May.

Figure 5 shows the occurrence rate of Es-region irregularities (EIr) and F-region irregularities (FIr) during the geomagnetic quiet period and the May 2024 storm. Under geomagnetic quiet conditions as shown in Figure 5a, EIr occurred mainly in the mid-latitudes of the Northern Hemisphere, with occurrence rates below 40%. Figures 5c and 5e present EIr and FIr during the May 2024 storm as obtained by RO observations. EIr increased markedly, with the maximum occurrence rate reaching 100% at high latitudes during the main phase and 50% at

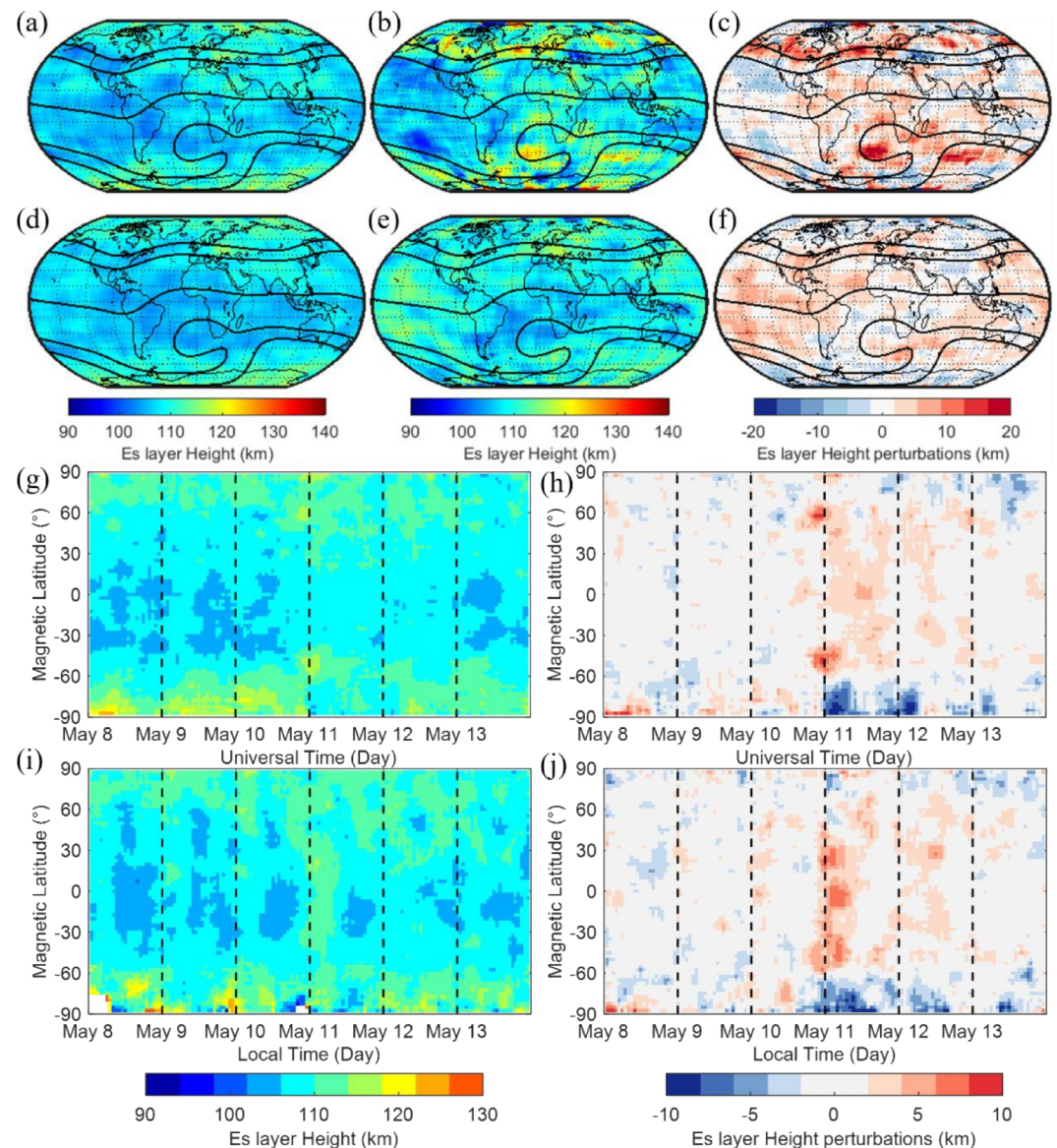


Figure 4. Same as Figure 3, but for EsH and EsHP.

mid- to low-latitude during the recovery phase. The spatial distributions of EIr were generally consistent with regions of enhanced EsI and EsP shown in Figure 3.

Figure 5 also presents FIr occurrence rates derived from RO observations. As shown in Figure 5a, FIr during quiet times was primarily confined to high latitudes. During the storm (Figures 5c and 5e), distinct enhancements were observed, with occurrence rates exceeding 100% at high latitudes during the main phase and 80% in the Australia–Asia and Southeast Pacific sectors during the recovery phase. In contrast, FIr occurrence rates at mid- to low-latitudes between 60°W and 60°E were relatively lower than quiet-time values. This suppression may be attributed to reduced background electron density. Since the development of irregularities depends on absolute fluctuations in ionospheric electron density (Chang et al., 2025a), a low background density can inhibit the detection of FIr even in the presence of substantial electron density variations.

FIr was also derived from ground-based GNSS observations for comparison. The results from ground-based GNSS observations show good agreement with RO-derived FIr distributions during both quiet and storm periods, including high-latitude enhancements and recovery-phase enhancements over the Australia–Asia sector. Meanwhile, the spatial coverage of FIr obtained by ground-based observations remains limited, despite increased

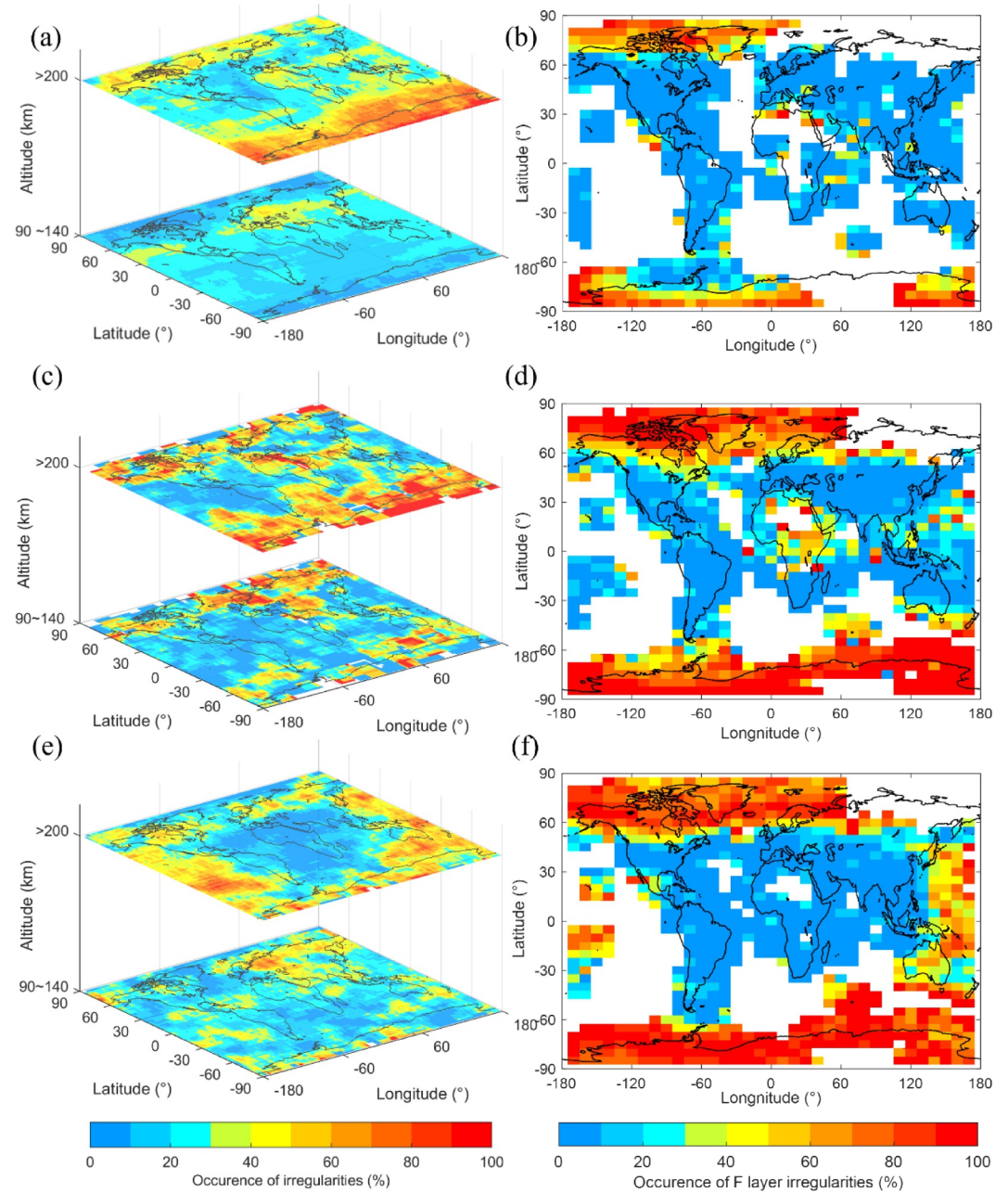


Figure 5. The 3-D mapping of EIr and FIr during the geomagnetic quiet times (a, b), the main phase (c, d) and recovery phase (e, f) of the May 2024 storm. (a, c, and e) EIr and FIr derived from RO observations. (b, d, and f) FIr obtained from ground-based GNSS stations. The main and recovery phases of the storm were defined as 17:00–24:00 UTC on 10 May and 10:00–20:00 UTC on 11 May, respectively. Data from the same local time intervals on 7–8 May and 14–15 May served as the quiet-time reference.

FIr also being detected over the Southeast Pacific during the quiet time and the storm main phase. The overall consistency between RO and ground-based FIr confirms the feasibility of using the RO technique for global FIr monitoring.

The occurrence rates of EIr and FIr were calculated based on the tangent point information from the RO profiles. Three occurrence ratios are defined, that is, EIr occurrence rate = $\frac{\text{Number of profiles with detected EIr}}{\text{Total number of RO profiles}}$, FIr occurrence rate = $\frac{\text{Number of profiles with detected FIr}}{\text{Total number of RO profiles}}$, and co-occurrence rate = $\frac{\text{Number of profiles with detected EIr and FIr concurrently}}{\text{Total number of RO profiles}}$.

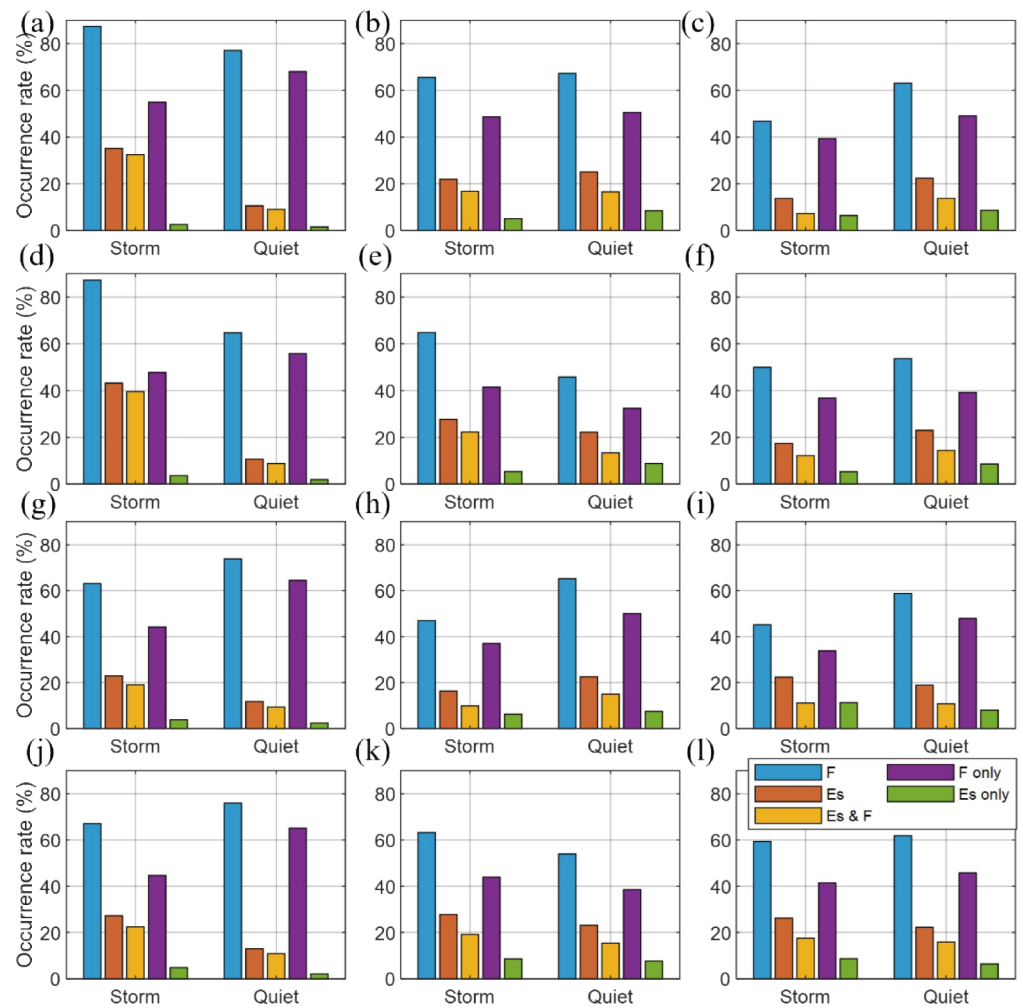


Figure 6. Occurrence rate of Elr, Flr, EIro, FIro, and the concurrence rate of Elr and Flr, during daytime main phase (Storm) and magnetic quiet time (Quiet) at high latitude, mid latitude, and low latitudes (a–c). Same as (a–c), but for nighttime (d–f), for recovery phase daytime (g–i), and nighttime (j–l). The daytime and nighttime were defined as 6–18 LT and 18–6 LT respectively. The occurrence rate of Elr, Flr, their concurrence rate, EIro, and FIro are denoted as F, Es, Es, F, and F only, Es only in legend.

The EIro-only occurrence rate (EIro) and the FIro-only occurrence rate (FIro) are defined as the occurrence rates for events where only EIro is present (without FIro) and where only FIro is present (without EIro), respectively.

Figure 6 presents the occurrence rates of EIro, FIro and the co-occurrence rates of EIro and FIro across high-, mid-, and low-MLAT during daytime and nighttime. At high latitudes, compared to geomagnetically quiet conditions, the co-occurrence rate showed a pronounced increase during both the main and recovery phases, specifically, it rose to 32% (daytime) and 40% (nighttime) during the main phase, more than four times the quiet-time values of 7% and 9%. At mid-latitudes, the co-occurrence rate increased at nighttime but decreased during daytime relative to quiet conditions. While at low latitudes, it increased during the recovery phase but decreased during the main phase.

Variations in EIro occurrence rates were consistent with those of the co-occurrence rate, whereas FIro showed a different pattern. FIro increased primarily at high latitudes during the main phase but was generally suppressed under other conditions. The extreme depletion of ionospheric electron density during the storm is likely responsible for the overall reduction in FIro occurrence.

At high latitudes, the occurrence rate of EIro remained nearly unchanged, while that of FIro even declined. The observed increases in EIro and FIro were thus primarily driven by the enhancement in their co-occurrence during

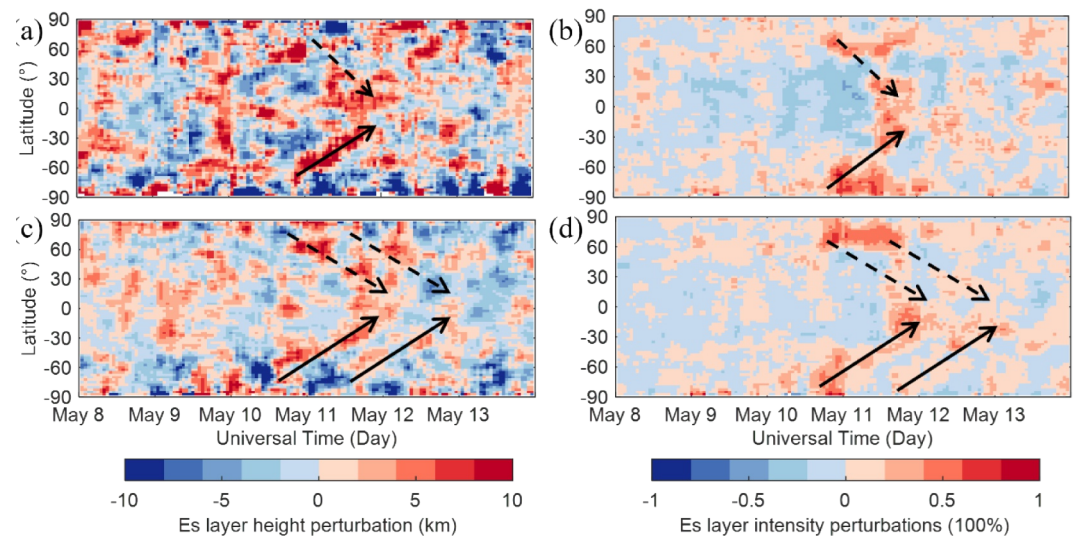


Figure 7. Time series of EsP (left panels) and EsHP (right panels) over the Southeast Pacific sectors ($-150^{\circ}\text{W} \sim -90^{\circ}\text{W}$) (a, b) and Australia-Asia sectors ($120^{\circ}\text{E}-180^{\circ}\text{E}$) (c), (d). The black arrows indicate potential wave-like propagation.

both the main and recovery phases. At mid-latitudes, the occurrence rate of Elro decreased, whereas FIro and FIR increased substantially during nighttime. Consequently, the rise in co-occurrence rate at mid-latitudes is likely attributable to the pronounced increase in FIR, which may further influence the generation of E-layer irregularities through E-F layer coupling processes (see detailed discussion in Section 4.2). At low latitudes, variations in the occurrence rates of Elro and FIro closely followed those of FIR and Elr, respectively.

3.2. Es-Layer Morphology Across Longitudinal Sectors

Figure 7 reveals a time-delayed (wave-like) propagation of enhanced Es layers that is likely to occur from the auroral zone toward the equatorial region in the Southern Hemisphere during the storm. This phenomenon is examined using the EsP and EsHP parameters over the Southeast Pacific and Australia-Asia sectors. As indicated by the black solid lines in Figures 7b and 7d, this wave-like propagation initiated in the auroral region around 19:00 UT on May 10, propagating at a speed of approximately 150 km per hour in the Southeast Pacific sector. A similar initiation time was observed in the Australia-Asia sector, with a higher propagation speed of about 300 km per hour. As shown in Figures 7a and 7c, these wave-like characteristics were also observed in the EsHP. In the Northern Hemisphere, however, similar time-delayed propagation of EsP and EsHP was unclear in these two regions.

Figure 8 presents the statistical occurrence rates of Elr, FIR, Elro, FIro and the co-occurrence rate of Elr and FIR over the Southeast Pacific and Australia-Asia sectors. The indices of Elr, FIR and their co-occurrence rate increased, specifically in the Southeast Pacific region during nighttime. The co-occurrence rate reached 22%, more than four times that observed under geomagnetically quiet conditions (5%). Meanwhile, the occurrence rates of Elr and FIR increased concurrently from 12% to 27% and from 40% to 85%, respectively. These enhancements are substantially higher than the global baseline increase, which is below 10% shown in Figure 6k.

Moderate increases were observed in the Southeast Pacific region during daytime, as well as in the Australia-Asia sector. The co-occurrence rate rose from 9% to 17% in the Southeast Pacific during daytime, and from 12% (16%) to 21% (23%) in the Australia-Asia sector during daytime (nighttime).

Compared to quiet conditions, the nighttime occurrence rate of FIro during the storm increased notably from 40% (42%) to 56% (47%) in the Southeast Pacific and Australia-Asia sectors, accompanied by a concurrent decrease in Elro. During daytime, the differences in occurrence rates between storm and quiet periods during the daytime were far smaller than those observed at nighttime for both FIro and Elro.

Figure 9 shows the time series of EsP and EsHP over longitudinal sectors of $180^{\circ}\text{W}-150^{\circ}\text{W}$, $90^{\circ}\text{W}-0^{\circ}\text{W}$, and $0^{\circ}\text{E}-90^{\circ}\text{E}$. In the Southern Hemisphere portion of the $0^{\circ}\text{E}-90^{\circ}\text{E}$ sector, a weak wave-like propagation can be

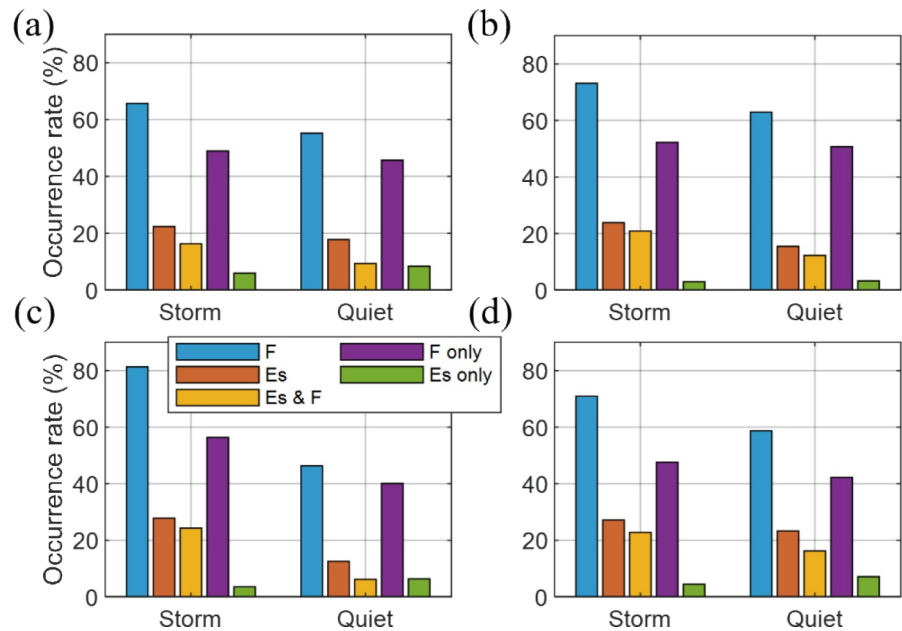


Figure 8. Occurrence rate of Elr, Fir, Elro, and Firo, and the concurrence rate of Elr and Fir, over Southeast Pacific region during daytime (a) and nighttime (c). In panels (b, d) same as (a, c), but for Australia-Asia sector.

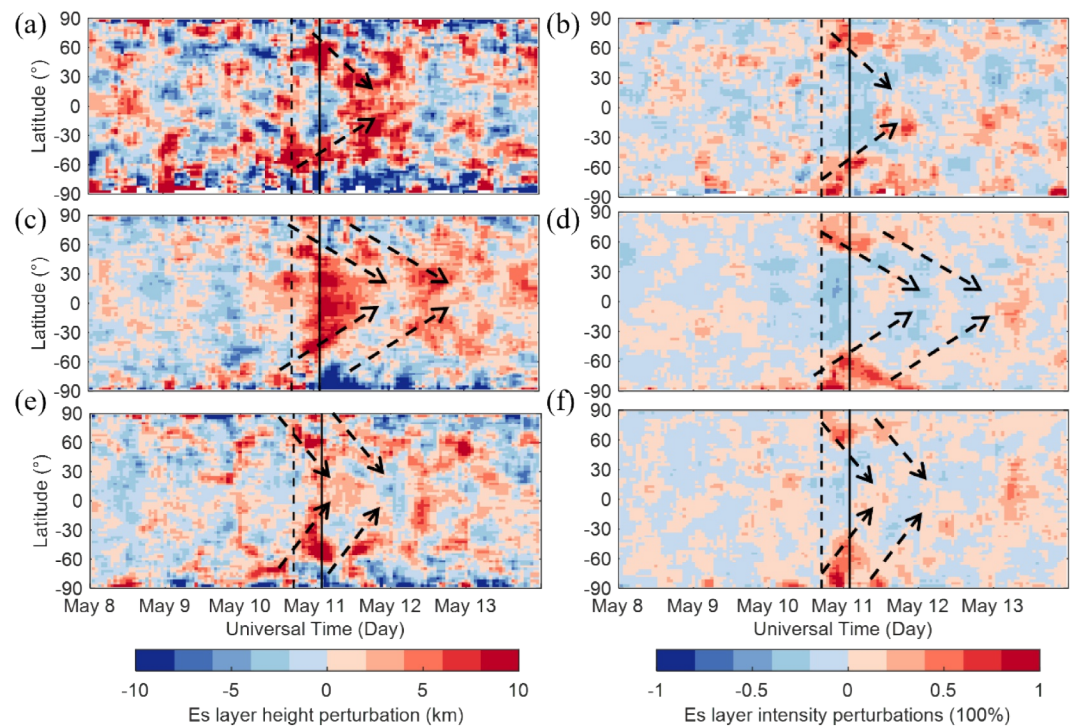


Figure 9. Shows time series of EsHP over (a) 180°W–150°W, (c) 90°W–0°W, and (e) 0°E–90°E, and the EsP for the same longitudinal sectors in (b, d, and f) respectively. The dashed line marks 17:00 UT on 11 May 2024, corresponding to the onset of the storm main phase. The solid line indicates 02:00 UT on 11 May 2024, representing the onset of the storm recovery phase. The black dashed arrows indicate potential wave-like propagation.

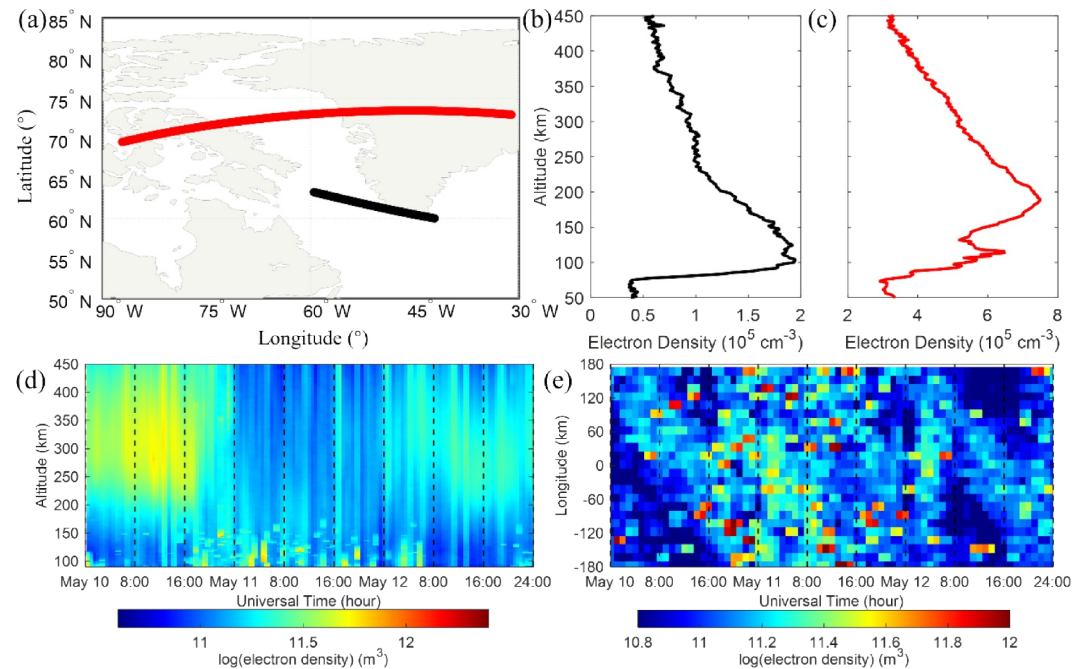


Figure 10. (a) Tangent point trajectories of the two RO events. The black curve denotes event C45-002, which occurred at 00:51 UT on 11 May, with signals transmitted by BeiDou PRN 45 (C45) and tracked by the Tianmu-1 002 satellite. The red curve denotes event C48-002, which occurred at 00:55 UT on 11 May, where signals were transmitted by BeiDou PRN 48 (C48) and tracked by the Tianmu-1 002 satellite. (b) Electron density profiles from event C45-002. (c) Electron density profiles from event C48-002. (d) Altimetrical variation of electron density in high-latitude regions as a function of universal time. (e) Longitudinal variation of electron density in high-latitude regions as a function of universal time.

observed, although it was significantly weaker than those in the Southeast Pacific and Australia-Asia regions. In the other sectors, no discernible wave-like propagation signatures were detected. The significant longitudinal variations of the Es layer response to the storm are primarily due to the geomagnetic pole position and local time dependence (Astafyeva et al., 2015).

4. Discussions

Over 25,000 RO profiles are analyzed to investigate global Es-layer disturbances during the May 2024 storm. The results reveal different behaviors in both EsI and EsH across latitudes and throughout the storm phases.

4.1. The Characteristics of High-Latitude EsI

The enhancement of EsI at high latitudes (above 60°MLAT) is likely associated with energetic particle precipitation in the auroral zone. Figure 10 displays two electron density profiles and the temporal variation of electron density in statistics. While the electron density typically peaks above 200 km, Figures 10b and 10c reveals a distinct peak near 100 km within the Es region during the storm main phase, and even in some cases, exhibits dual peaks at approximately 190 and 100 km. This E-region-dominated ionosphere has been linked to polar particle precipitation in previous studies (Mayer and Jakowski, 2009; Chang, Morton, et al., 2025).

Figure 10d shows the temporal evolution of average electron density over high-latitude regions with 1-hr and 5-km bins. Prior to the storm, the electron density peaked at F-region altitudes between 200 and 400 km. With the onset of the storm at 17:00 UT, a pronounced depletion of F-region electron density was accompanied by a marked enhancement in the Es region. Chang, Morton, et al. (2025) reported an E-region ionospheric enhancement from Poker Flat incoherent-scatter radar observations, however, the corresponding signature in RO profiles was less distinct mainly owing to the relatively limited number of profiles. Based on an unprecedented number of RO profiles with comprehensive longitudinal coverage, this study clearly reveals the large-scale occurrence of an E-layer-dominated ionosphere across the high-latitude region.

As shown in Figure 10e, the E-region electron density reached values up to 10^{11} m^{-3} , with a clear longitudinal dependence due to local time variations before 16:00 UT on 10 May and after 08:00 UT on 12 May. During the storm, E-region electron densities attained 10^{12} m^{-3} with unclear longitudinal dependence. The appearance of an E-region-dominated ionosphere over all longitudes indicated that the enhanced EsI at high latitudes may be primarily driven by energetic particle precipitation within the auroral zone.

RO observations at high latitudes reveal a simultaneous enhancement of EIr, FIr, and their co-occurrence rates during the storm main phase. Intense particle precipitation may not only create an E-region-dominated ionosphere but also induce sharp gradients in electron density that produce EIr. Hu et al. (2025) demonstrated that enhanced polar energy input raised electron temperatures in the F region to as high as 4000 K. Such energy inputs further drive electron-density gradients and turbulences (Chen et al., 2025; Foster et al., 2024; Fu et al., 2025; Jin et al., 2024; Themens et al., 2024; Wang et al., 2025), which likely increased FIr shown in this study.

4.2. Characteristics of Mid-to-Low-Latitude Es Layer During the Recovery Phase

EsI enhancement was observed at mid- and low-latitudes during the storm recovery phase, particularly over the Southeast Pacific and Australia-Asia sectors. These enhancements exhibited wave-like equatorward propagation in the Southern Hemisphere, accompanied by an approximately 10 km rise in EsH. Based on simulations, Qiu et al. (2025b) identified strengthened westward wind as the primary driver of vertical ion convergence (VIC). As a key parameter governing Es-layer formation (Naufal et al., 2025), VIC also showed similar wave-like propagations. The RO data in this study provide additional observational evidence that disturbed neutral winds may play a significant role in Es layer generation in the Southern Hemisphere.

The enhanced co-occurrence of EIr and FIr irregularities indicates strengthened interlayer coupling which was pronounced at night. The increases in the occurrence rates of FIr indicate highly disturbed state of the F-region ionosphere during the storm. Thus, disturbance dynamo effects in the F-region likely served as an additional driver of the EsI enhancement. Qiu et al. (2025b) shows that the equatorial VIC can also be significantly intensified during this storm when only considering electric field effects in simulations.

In the Northern Hemisphere, however, RO observations show that the wave-like propagation is very weak for both EsP and EsHP (indicated by dashed arrows in Figure 6). Qiu and Lin. (2025a) show weaker propagation signals in the Northern Hemisphere compared to those observed in the Southern Hemisphere based on ionosonde measurements. The spatial structure of the possible propagation remains uncertain due to the limitations of sparse ionosondes measurements. The lack of a distinct wave-like propagation in our RO data suggests that Es-layer formation in the Northern Hemisphere during this event may be modulated by mechanisms beyond neutral winds. The hemispheric asymmetries in storm-time energy deposition, the ratio of atomic oxygen column density to molecular nitrogen column density ($\Sigma\text{O}/\text{N}_2$), the development of equatorial plasma bubbles (EPBs) as well as the seasonal variability of the Es layer itself may contribute to the observed E-region hemispheric disparity (Evans et al., 2024; Hu et al., 2025; Yang et al., 2025). The relative contribution of each driver warrants further investigation through ionosphere-thermosphere models.

4.3. Characteristics of Mid-to-Low-Latitude Es Layer During the Main Phase

Interestingly, EsI depletion was observed at mid- and low magnetic latitudes during the main phase. Figure 11 shows the temporal variation of EsP together with the Interplanetary Electric Field (Ey). From 17:00 UT on May 10 to 03:00 UT on May 11, Ey reached a maximum of 45 mV/m, indicating an intense eastward Prompt Penetration Electric Field (PPEF) event (Ross & Huang, 2021). This period coincided with a sharp EsI reduction exceeding 50% in the Northern Hemisphere. In theory, such a strong eastward PPEF can facilitate rapid vertical transport of plasma away from the E region, thereby depleting EsI regardless of the presence of disturbed neutral winds (Abdu et al., 2007, 2013; Resende et al., 2013). The intense eastward PPEF, which drove vertical $\mathbf{E} \times \mathbf{B}$ drifts above 95 m/s during the main phase, has also been reported by Astafyeva et al. (2025), although their analysis focused primarily on the impact on the F-region ionosphere.

After about 03:30 UT on 11 May, the PPEF subsided, and the EsI gradually recovered and exhibited a clear enhancement during the recovery phase. Thus, we conjecture that the intense PPEF may be the driver for the depletion of EsI at mid- and low-latitudes during the storm main phase.

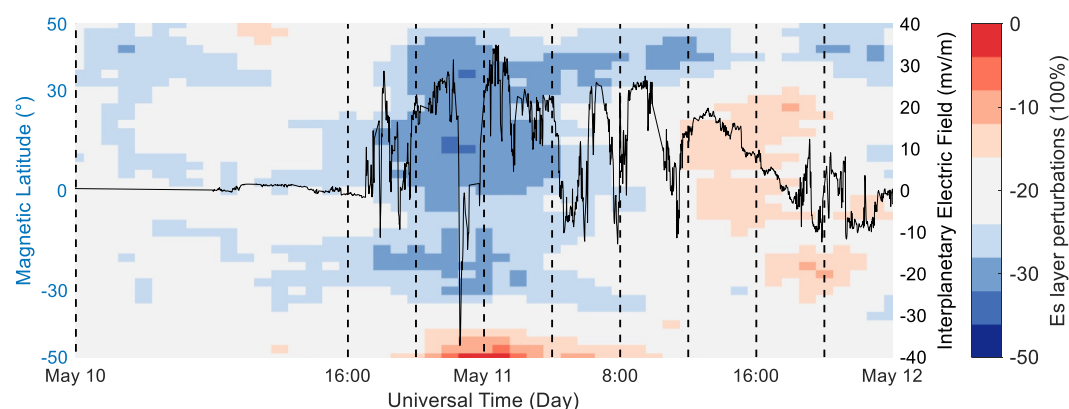


Figure 11. Geographical and temporal distributions of EsP derived from RO observations and the temporal distributions of Interplanetary Electric Field (Ey).

5. Conclusions

This study presents a global analysis of the Es layer response to the May 2024 storm by utilizing an unprecedented data set of more than 25,000 daily RO events. Observations revealed that the storm caused extensive disturbances in the Es region across the globe. The major findings are summarized as follows:

Following the storm onset, the Es layer exhibited rapid global responses and notable expansions during different phases of the May 2024 storm, revealing the latitudinal variations and hemispheric (North-South) asymmetry of the Es-layer disturbances.

There were distinctive impact responses at different latitudinal sectors. At high latitudes, significant enhancements in Es-layer intensity, accompanying by notable Es-layer height elevations, were recorded throughout the storm. The peak Es-layer intensity reached approximately four times the values during geomagnetic quiet conditions, while Es-layer height increased by up to 20 km. In contrast, at mid- and low-latitude regions, Es-layer generally exhibit depletion during the main phase and enhancement during the recovery phase.

The Es layer over the Southeast Pacific exhibited complex disturbances and associated generation mechanisms. Observations revealed potentially enhanced coupling between E- and F-region ionosphere, as evidenced by a fourfold increase in the simultaneous occurrence rate of E- and F-region irregularities compared to quiet periods. This enhanced interlayer coupling was predominantly nocturnal, suggesting the possible downward influence from the F-region to the E-region. Furthermore, Es-layer perturbations displayed clear wave-like propagation from polar to equatorial regions in the Southern Hemisphere. As previous studies have demonstrated similar propagation characteristics for VIC in simulations, our study provides additional observational evidence that disturbed neutral wind may also serve as a key driver for Es-layer formation.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Calculation of geomagnetic field lines were performed using the apexpy python package and can be downloaded from <https://github.com/aburrell/apexpy>. COSMIC-2 RO data are available through the UCAR COSMIC Data Analysis and Archive Center (CDAAC) website: <https://cdaac-www.cosmic.ucar.edu/cdaac/products.html>. Tianmu-1 RO data sets in the present study are available in Guo (2025c) (<https://doi.org/10.5281/zenodo.17489554>). Data from ground-based GNSS stations are available from UCSD <https://garner.ucsd.edu/>, CORS (<https://geodesy.noaa.gov/CORS/data.shtml>), IGS (<https://www.earthdata.nasa.gov/data/space-geodesy-techniques/gnss/daily-30-second-data-product>).

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